Mode 6 — Square (centered / opt)

DP32 (8×8) → DP16 (8×8) → DP16 (8×4) → DP8 (8×4) → DP8 (4×4) → centered add-and-square

Step 1 — DP32 (8×8)

$$\mathrm{DP32}\,(8\times8) = \sum_{i=0}^{31} a_i\, b_i$$

Step 2 — DP16 (8×8)

$$\mathrm{DP32}\,(8\times8) = \underbrace{\sum_{i=0}^{15} a_i\, b_i}_{\mathrm{DP16}\,(8\times8)} + \underbrace{\sum_{i=16}^{31} a_i\, b_i}_{\mathrm{DP16}\,(8\times8)}$$

Step 3 — DP16 (8×4)

$$\mathrm{DP32}\,(8\times8) = 2^4 \underbrace{\sum_{i=0}^{15} a_i\, b_i^{hi}}_{\mathrm{DP16}\,(8\times4)} + \underbrace{\sum_{i=0}^{15} a_i\, b_i^{lo}}_{\mathrm{DP16}\,(8\times4)} + 2^4 \underbrace{\sum_{i=16}^{31} a_i\, b_i^{hi}}_{\mathrm{DP16}\,(8\times4)} + \underbrace{\sum_{i=16}^{31} a_i\, b_i^{lo}}_{\mathrm{DP16}\,(8\times4)}$$

Step 4 — DP8 (8×4)

$$\begin{aligned}\mathrm{DP32}\,(8\times8) = {} & 2^4 \underbrace{\sum_{i=0}^{7} a_i\, b_i^{hi}}_{\mathrm{DP8}\,(8\times4)} + 2^4 \underbrace{\sum_{i=8}^{15} a_i\, b_i^{hi}}_{\mathrm{DP8}\,(8\times4)} + \underbrace{\sum_{i=0}^{7} a_i\, b_i^{lo}}_{\mathrm{DP8}\,(8\times4)} + \underbrace{\sum_{i=8}^{15} a_i\, b_i^{lo}}_{\mathrm{DP8}\,(8\times4)} \\ & + 2^4 \underbrace{\sum_{i=16}^{23} a_i\, b_i^{hi}}_{\mathrm{DP8}\,(8\times4)} + 2^4 \underbrace{\sum_{i=24}^{31} a_i\, b_i^{hi}}_{\mathrm{DP8}\,(8\times4)} + \underbrace{\sum_{i=16}^{23} a_i\, b_i^{lo}}_{\mathrm{DP8}\,(8\times4)} + \underbrace{\sum_{i=24}^{31} a_i\, b_i^{lo}}_{\mathrm{DP8}\,(8\times4)}\end{aligned}$$

Step 5 — DP8 (4×4)

$$\begin{aligned}\mathrm{DP32}\,(8\times8) = {} & 2^8 \underbrace{\sum_{i=0}^{7} a_{H,i}\, b_i^{hi}}_{\mathrm{DP8}\,(4\times4)} + 2^4 \underbrace{\sum_{i=0}^{7} a_{L,i}\, b_i^{hi}}_{\mathrm{DP8}\,(4\times4)} + 2^8 \underbrace{\sum_{i=8}^{15} a_{H,i}\, b_i^{hi}}_{\mathrm{DP8}\,(4\times4)} + 2^4 \underbrace{\sum_{i=8}^{15} a_{L,i}\, b_i^{hi}}_{\mathrm{DP8}\,(4\times4)} \\ & + 2^4 \underbrace{\sum_{i=0}^{7} a_{H,i}\, b_i^{lo}}_{\mathrm{DP8}\,(4\times4)} + \underbrace{\sum_{i=0}^{7} a_{L,i}\, b_i^{lo}}_{\mathrm{DP8}\,(4\times4)} + 2^4 \underbrace{\sum_{i=8}^{15} a_{H,i}\, b_i^{lo}}_{\mathrm{DP8}\,(4\times4)} + \underbrace{\sum_{i=8}^{15} a_{L,i}\, b_i^{lo}}_{\mathrm{DP8}\,(4\times4)} \\ & + 2^8 \underbrace{\sum_{i=16}^{23} a_{H,i}\, b_i^{hi}}_{\mathrm{DP8}\,(4\times4)} + 2^4 \underbrace{\sum_{i=16}^{23} a_{L,i}\, b_i^{hi}}_{\mathrm{DP8}\,(4\times4)} + 2^8 \underbrace{\sum_{i=24}^{31} a_{H,i}\, b_i^{hi}}_{\mathrm{DP8}\,(4\times4)} + 2^4 \underbrace{\sum_{i=24}^{31} a_{L,i}\, b_i^{hi}}_{\mathrm{DP8}\,(4\times4)} \\ & + 2^4 \underbrace{\sum_{i=16}^{23} a_{H,i}\, b_i^{lo}}_{\mathrm{DP8}\,(4\times4)} + \underbrace{\sum_{i=16}^{23} a_{L,i}\, b_i^{lo}}_{\mathrm{DP8}\,(4\times4)} + 2^4 \underbrace{\sum_{i=24}^{31} a_{H,i}\, b_i^{lo}}_{\mathrm{DP8}\,(4\times4)} + \underbrace{\sum_{i=24}^{31} a_{L,i}\, b_i^{lo}}_{\mathrm{DP8}\,(4\times4)}\end{aligned}$$

Step 6 — Centered add-and-square (subtract 8 from unsigned nibbles a_L, b^{lo}; per DP8 (4×4) add $c = 0/512/2048$ for 0/1/2 unsigned nibbles)

$$\mathrm{DP32}\,(8\times8) = 2^8 \underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7}(a_{H,i}+b_i^{hi})^2 - \sum_{i=0}^{7} a_{H,i}^2 - \sum_{i=0}^{7}(b_i^{hi})^2\Big)}_{\mathrm{DP8}\,(4\times4)}$$

$$+ 2^4 \underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7} \big((a_{L,i}-8)+b_i^{hi}\big)^2 - \sum_{i=0}^{7}(a_{L,i}-8)^2 - \sum_{i=0}^{7}(b_i^{hi}-8)^2 + 512\Big)}_{\text{DP8 (4×4)}}$$

$$+ 2^8 \underbrace{\tfrac{1}{2}\Big(\sum_{i=8}^{15}(a_{H,i}+b_i^{hi})^2 - \sum_{i=8}^{15} a_{H,i}^2 - \sum_{i=8}^{15}(b_i^{hi})^2\Big)}_{\text{DP8 (4×4)}}$$

$$+ 2^4 \underbrace{\tfrac{1}{2}\Big(\sum_{i=8}^{15} \big((a_{L,i}-8)+b_i^{hi}\big)^2 - \sum_{i=8}^{15}(a_{L,i}-8)^2 - \sum_{i=8}^{15}(b_i^{hi}-8)^2 + 512\Big)}_{\text{DP8 (4×4)}}$$

$$+ 2^4 \underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7} \big(a_{H,i}+(b_i^{lo}-8)\big)^2 - \sum_{i=0}^{7}(a_{H,i}-8)^2 - \sum_{i=0}^{7}(b_i^{lo}-8)^2 + 512\Big)}_{\text{DP8 (4×4)}}$$

$$+ \underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7} \big((a_{L,i}-8)+(b_i^{lo}-8)\big)^2 - \sum_{i=0}^{7}(a_{L,i}-16)^2 - \sum_{i=0}^{7}(b_i^{lo}-16)^2 + 2048\Big)}_{\text{DP8 (4×4)}}$$

$$+ 2^4 \underbrace{\tfrac{1}{2}\Big(\sum_{i=8}^{15} \big(a_{H,i}+(b_i^{lo}-8)\big)^2 - \sum_{i=8}^{15}(a_{H,i}-8)^2 - \sum_{i=8}^{15}(b_i^{lo}-8)^2 + 512\Big)}_{\text{DP8 (4×4)}}$$

$$+ \underbrace{\tfrac{1}{2}\Big(\sum_{i=8}^{15} \big((a_{L,i}-8)+(b_i^{lo}-8)\big)^2 - \sum_{i=8}^{15}(a_{L,i}-16)^2 - \sum_{i=8}^{15}(b_i^{lo}-16)^2 + 2048\Big)}_{\text{DP8 (4×4)}}$$

$$+ 2^8 \underbrace{\tfrac{1}{2}\Big(\sum_{i=16}^{23}(a_{H,i}+b_i^{hi})^2 - \sum_{i=16}^{23} a_{H,i}^2 - \sum_{i=16}^{23}(b_i^{hi})^2\Big)}_{\text{DP8 (4×4)}}$$

$$+ 2^4 \underbrace{\tfrac{1}{2}\Big(\sum_{i=16}^{23} \big((a_{L,i}-8)+b_i^{hi}\big)^2 - \sum_{i=16}^{23}(a_{L,i}-8)^2 - \sum_{i=16}^{23}(b_i^{hi}-8)^2 + 512\Big)}_{\text{DP8 (4×4)}}$$

$$+ 2^8 \underbrace{\tfrac{1}{2}\Big(\sum_{i=24}^{31}(a_{H,i}+b_i^{hi})^2 - \sum_{i=24}^{31} a_{H,i}^2 - \sum_{i=24}^{31}(b_i^{hi})^2\Big)}_{\text{DP8 (4×4)}}$$

$$+ 2^4 \underbrace{\tfrac{1}{2}\Big(\sum_{i=24}^{31} \big((a_{L,i}-8)+b_i^{hi}\big)^2 - \sum_{i=24}^{31}(a_{L,i}-8)^2 - \sum_{i=24}^{31}(b_i^{hi}-8)^2 + 512\Big)}_{\text{DP8 (4×4)}}$$

$$+ 2^4 \underbrace{\tfrac{1}{2}\Big(\sum_{i=16}^{23} \big(a_{H,i}+(b_i^{lo}-8)\big)^2 - \sum_{i=16}^{23}(a_{H,i}-8)^2 - \sum_{i=16}^{23}(b_i^{lo}-8)^2 + 512\Big)}_{\text{DP8 (4×4)}}$$

$$+ \underbrace{\tfrac{1}{2}\Big(\sum_{i=16}^{23} \big((a_{L,i}-8)+(b_i^{lo}-8)\big)^2 - \sum_{i=16}^{23}(a_{L,i}-16)^2 - \sum_{i=16}^{23}(b_i^{lo}-16)^2 + 2048\Big)}_{\text{DP8 (4×4)}}$$

$$+\,\underbrace{2^4\,\tfrac{1}{2}\Big(\sum_{i=24}^{31}\big(a_{H,i}+(b_i^{lo}-8)\big)^2-\sum_{i=24}^{31}(a_{H,i}-8)^2-\sum_{i=24}^{31}(b_i^{lo}-8)^2+512\Big)}_{\text{DP8 (4×4)}}$$

$$+\,\underbrace{\tfrac{1}{2}\Big(\sum_{i=24}^{31}\big((a_{L,i}-8)+(b_i^{lo}-8)\big)^2-\sum_{i=24}^{31}(a_{L,i}-16)^2-\sum_{i=24}^{31}(b_i^{lo}-16)^2+2048\Big)}_{\text{DP8 (4×4)}}$$

Step 7 — All components

$$\begin{aligned}
\mathrm{PE} = {} & 2^8\sum_{i=0}^{7}(a_{H,i}+b_i^{hi})^2+2^4\sum_{i=0}^{7}\big((a_{L,i}-8)+b_i^{hi}\big)^2+2^8\sum_{i=8}^{15}(a_{H,i}+b_i^{hi})^2+2^4\sum_{i=8}^{15}\big((a_{L,i}-8)+b_i^{hi}\big)^2\\
&+2^4\sum_{i=0}^{7}\big(a_{H,i}+(b_i^{lo}-8)\big)^2+\sum_{i=0}^{7}\big((a_{L,i}-8)+(b_i^{lo}-8)\big)^2+2^4\sum_{i=8}^{15}\big(a_{H,i}+(b_i^{lo}-8)\big)^2+\sum_{i=8}^{15}\big((a_{L,i}-8)+(b_i^{lo}-8)\big)^2\\
&+2^8\sum_{i=16}^{23}(a_{H,i}+b_i^{hi})^2+2^4\sum_{i=16}^{23}\big((a_{L,i}-8)+b_i^{hi}\big)^2+2^8\sum_{i=24}^{31}(a_{H,i}+b_i^{hi})^2+2^4\sum_{i=24}^{31}\big((a_{L,i}-8)+b_i^{hi}\big)^2\\
&+2^4\sum_{i=16}^{23}\big(a_{H,i}+(b_i^{lo}-8)\big)^2+\sum_{i=16}^{23}\big((a_{L,i}-8)+(b_i^{lo}-8)\big)^2+2^4\sum_{i=24}^{31}\big(a_{H,i}+(b_i^{lo}-8)\big)^2+\sum_{i=24}^{31}\big((a_{L,i}-8)+(b_i^{lo}-8)\big)^2
\end{aligned}$$

$$\begin{aligned}
\alpha = {} & 2^8\sum_{i=0}^{7}a_{H,i}^2+2^4\sum_{i=0}^{7}(a_{L,i}-8)^2+2^8\sum_{i=8}^{15}a_{H,i}^2+2^4\sum_{i=8}^{15}(a_{L,i}-8)^2\\
&+2^4\sum_{i=0}^{7}(a_{H,i}-8)^2+\sum_{i=0}^{7}(a_{L,i}-16)^2+2^4\sum_{i=8}^{15}(a_{H,i}-8)^2+\sum_{i=8}^{15}(a_{L,i}-16)^2\\
&+2^8\sum_{i=16}^{23}a_{H,i}^2+2^4\sum_{i=16}^{23}(a_{L,i}-8)^2+2^8\sum_{i=24}^{31}a_{H,i}^2+2^4\sum_{i=24}^{31}(a_{L,i}-8)^2\\
&+2^4\sum_{i=16}^{23}(a_{H,i}-8)^2+\sum_{i=16}^{23}(a_{L,i}-16)^2+2^4\sum_{i=24}^{31}(a_{H,i}-8)^2+\sum_{i=24}^{31}(a_{L,i}-16)^2
\end{aligned}$$

$$\begin{aligned}
\beta = {} & 2^8\sum_{i=0}^{7}(b_i^{hi})^2+2^4\sum_{i=0}^{7}(b_i^{hi}-8)^2+2^8\sum_{i=8}^{15}(b_i^{hi})^2+2^4\sum_{i=8}^{15}(b_i^{hi}-8)^2\\
&+2^4\sum_{i=0}^{7}(b_i^{lo}-8)^2+\sum_{i=0}^{7}(b_i^{lo}-16)^2+2^4\sum_{i=8}^{15}(b_i^{lo}-8)^2+\sum_{i=8}^{15}(b_i^{lo}-16)^2\\
&+2^8\sum_{i=16}^{23}(b_i^{hi})^2+2^4\sum_{i=16}^{23}(b_i^{hi}-8)^2+2^8\sum_{i=24}^{31}(b_i^{hi})^2+2^4\sum_{i=24}^{31}(b_i^{hi}-8)^2\\
&+2^4\sum_{i=16}^{23}(b_i^{lo}-8)^2+\sum_{i=16}^{23}(b_i^{lo}-16)^2+2^4\sum_{i=24}^{31}(b_i^{lo}-8)^2+\sum_{i=24}^{31}(b_i^{lo}-16)^2
\end{aligned}$$

$$\begin{aligned}
C = {} & 0+8192+0+8192+8192+2048+8192+2048\\
&+0+8192+0+8192+8192+2048+8192+2048=73728
\end{aligned}$$